Experiment No:12
Experiment to detect outliers in a given data set.

AIM:
To detect outliers in a given data set.

DESCRIPTON:
Understand the procedure to identify the outliers in a given dataset

CODE:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Sample calculation for low range (lr), upper range (ur),
percentile

df = pd.read_csv("Student_Marks.csv")
array = df["Marks"].dropna().to_numpy()

array

array.mean()

np.percentile(array, 25)

np.percentile(array, 50)

np.percentile(array, 75)

np.percentile(array, 100)
```

```
# Outliers detection

def outDetection(array):
    sorted(array)
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)
lr, ur

# First diagram
sns.displot(array)
plt.show()

# Second diagram
sns.histplot(array, kde=True)
plt.show()

# Remove outliers
new_array = array[(array > lr) & (array < ur)]
new_array

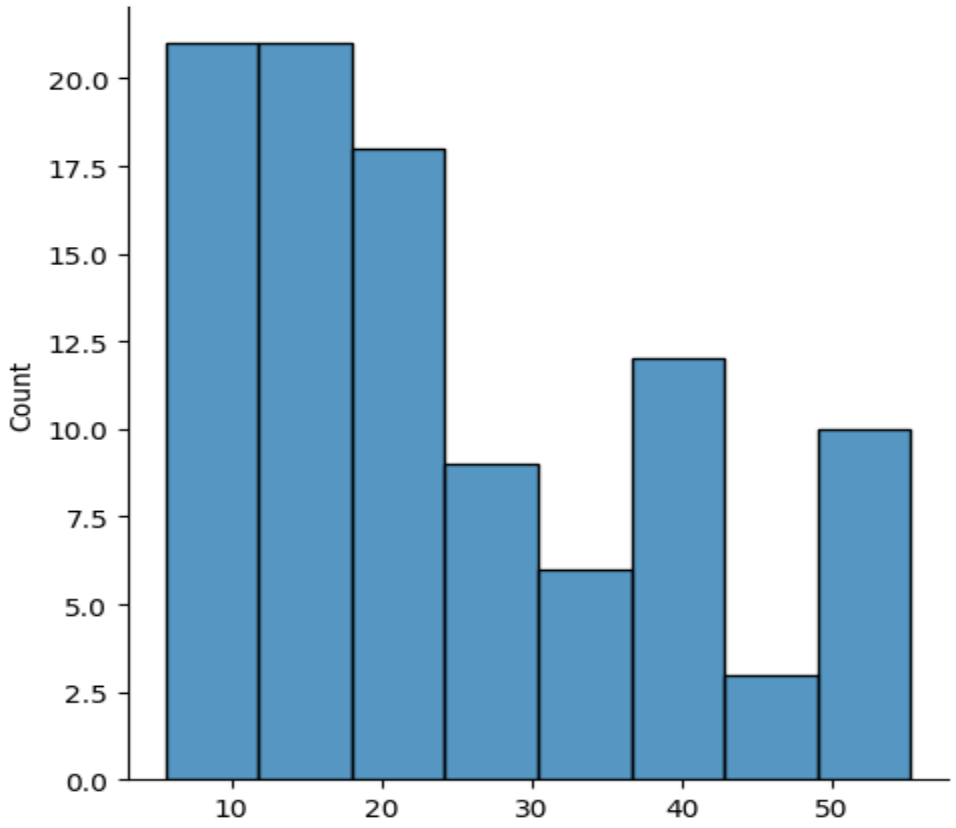
# Third diagram
sns.displot(new_array)
plt.show()

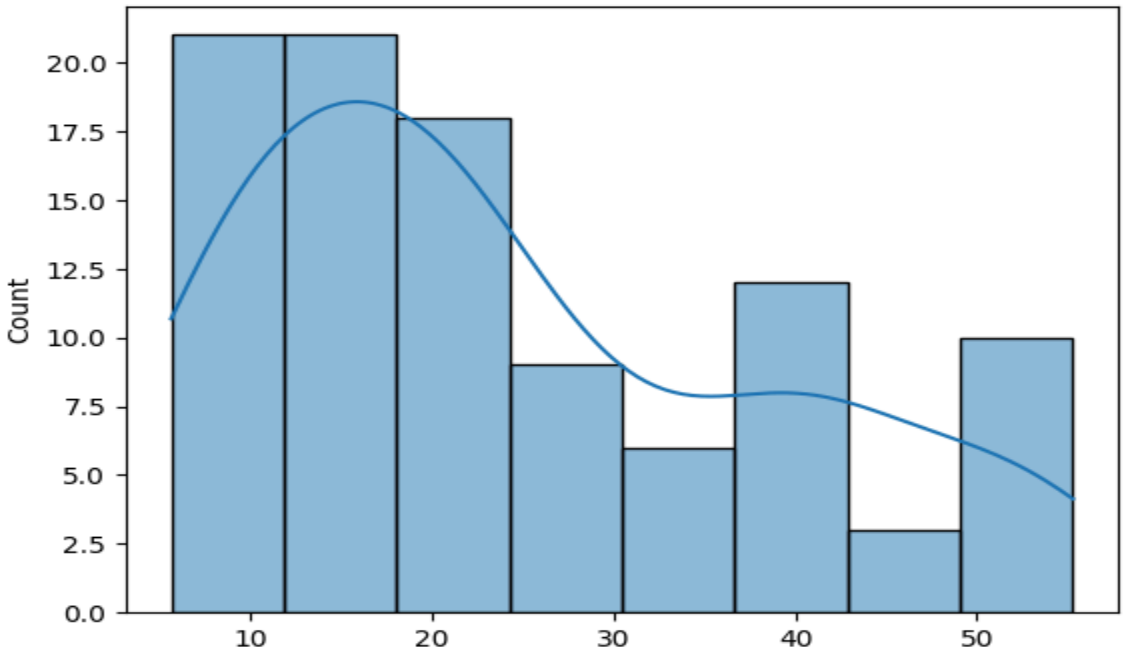
# Calculate new lower and upper ranges
lr1, ur1 = outDetection(new_array)
lr1, ur1
```

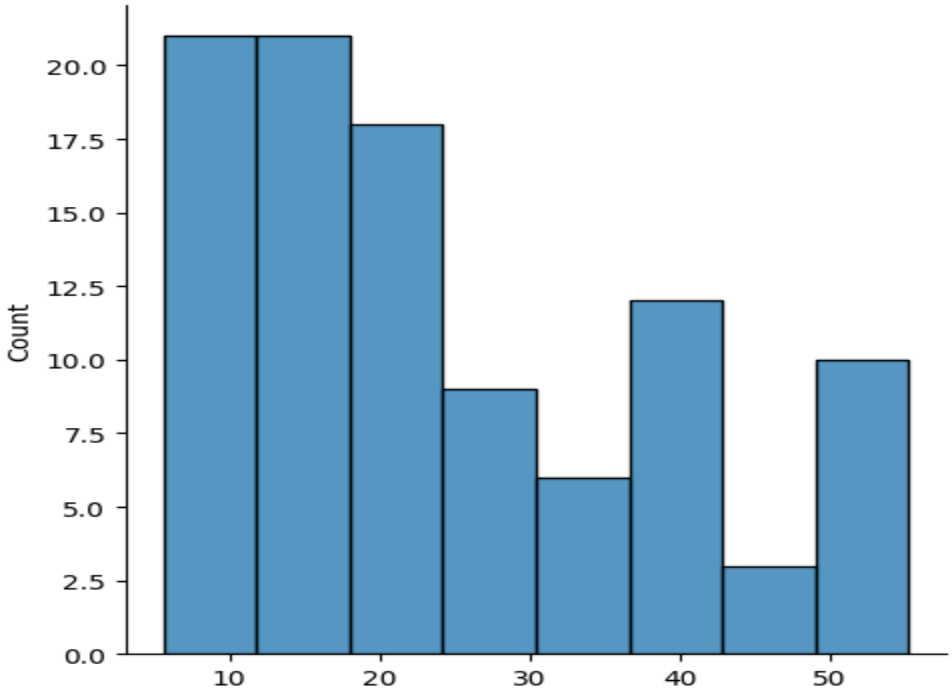
```
# Remove outliers again
final_array = new_array[(new_array > lr1) & (new_array < ur1)]
final_array

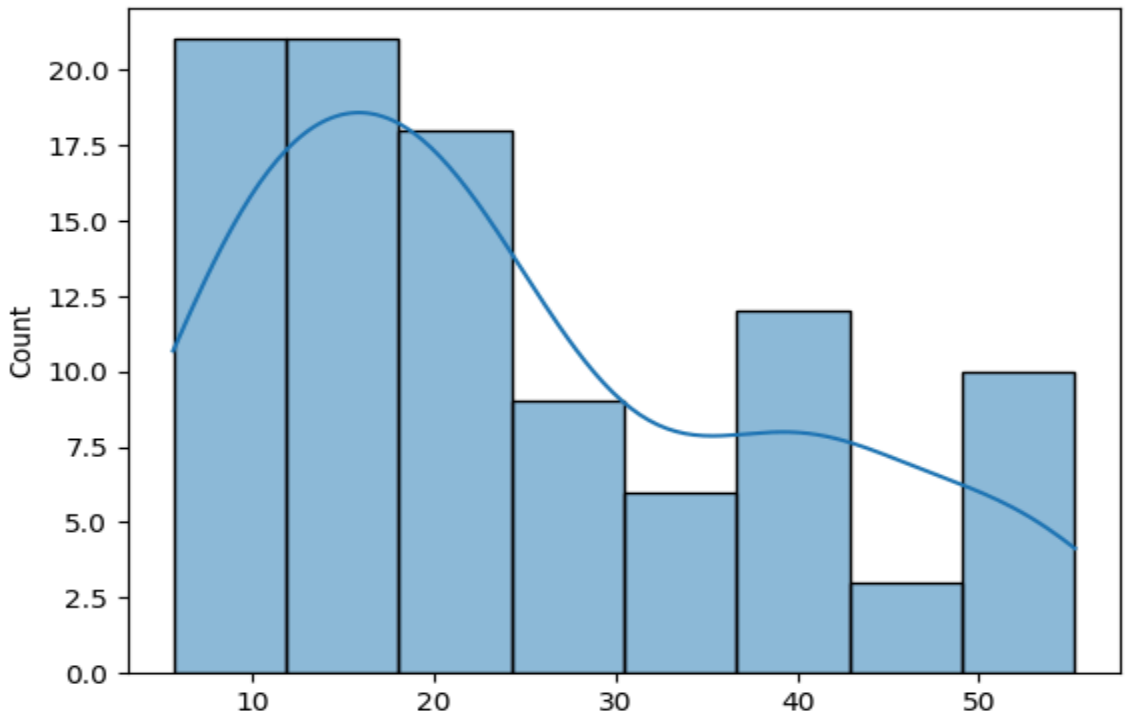
# Fourth diagram
sns.histplot(final_array, kde=True)
plt.show()
```

OUTPUT:




Count
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
10
20
30
40
50


Count
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
10
20
30
40
50



RESULT:

Thus, outliers in the given dataset were successfully detected using the Interquartile Range (IQR) method. The lower and upper limits were calculated using the first and third quartiles, and outliers were removed from the dataset. The distributions before and after removing outliers were visualized using Seaborn plots.